

Here's an outline of the topics covered in your "MAD repetition for test II," organized for clarity and study:

MAD repetition for Test II

I. Proof by Induction

This section tests your ability to apply the principle of mathematical induction to prove various statements (equations and inequalities).

- **Sums/Series:**

- - 1. $\sum_{i=1}^n (2i-1) = n^2$
- - 3. $\sum_{i=1}^n (3i-1) = (\sum_{i=1}^n ni) + n^2$

- **Inequalities:**

- - 2. $n^2 + 1 < 2n - 1$
- - 4. $4n^2 > n + 11$

II. Set Theory and Cardinality

This section assesses your understanding of different types of infinite sets and how to compare their sizes (cardinalities) using bijections or established theorems.

- **Identifying Cardinalities:**

- - 5. A: Set of negative integers
- - 6. B: The interval $(-1, 1)$
- - 7. C: Set of odd integers
- - 8. D: Power set of $\mathbb{N}$
- - 9. E: Set of real solutions to $\sin x = 1/2$
- - 10. F: Set of real numbers greater than 7
- - 11. G: Union of A and B ($A \cup B$)
- - 12. H: Cartesian product $\mathbb{Q} \times \mathbb{Q}$
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13. I: Set of natural numbers which are solutions of $x^2+6x+8=0$

- **Cartesian Products:**

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14. Cartesian product of $\{a,b,c\} \times \mathbb{Z}$

III. Functions

This section covers definitions of different types of functions, finding images and preimages, and inverse functions.

- **Function Definitions (Injectivity, Surjectivity, Bijectivity):**

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15. Define a surjection $f:\{5,7,9,0\} \rightarrow \mathbb{R}$ (and understanding why it's impossible).

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16. Define an injection $f:\mathbb{N} \rightarrow (0,1)$.

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17. Define a bijection $f:\mathbb{N} \rightarrow \{n \in \mathbb{N} : 5|n\}$ (and understanding why it's impossible for the standard interpretation, or providing one if the interpretation is different).

- **Image and Preimage of a Function:**

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18. Given $f:\mathbb{Z} \rightarrow \mathbb{R}$ with $f(x)=(x^2+1)/2$, find $f(A)$ for $A=\{3,5,6\}$ and $f^{-1}(B)$ for $B=\{1,5/8,0\}$.

- **Composition and Inverse Functions:**

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19. Given $f(x)=x+1$ and $g(x)=x-2$, find the inverse function of the composition $g \circ f$ or explain why it doesn't exist.

IV. Combinatorics

This section deals with counting principles for functions and relations.

- **Counting Functions:**

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20. How many functions can be defined from the set $\mathbb{N}$ to the set $\{a,b,c\}$?

- **Counting Relations:**

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21. How many relations $R \subseteq A \times B \times C$ can be defined if $|A|=4$, $|B|=2$, and $|C|=3$? (Note: This is a ternary relation, not a binary one).

V. Probability

This section focuses on calculating probabilities, including conditional probability and checking for independence of events.

- **Events and Independence:**

- - 22. From the set of natural numbers bigger than 10 and smaller than or equal to 20, select 3 numbers.
 - A: Event of obtaining an even sum.
 - B: Event of obtaining at least one odd number.
 - Determine if A and B are independent events.
 - Find $P(A|B)$.
- **Set Theory in Probability (Inclusion-Exclusion Principle):**
 - - 23. Among 20 students, everyone speaks Italian or English. 7 speak Italian, 17 speak English. What is the probability that a randomly selected student speaks both languages?
 - - 24. From the set of natural numbers between 100 and 460, select one number. How to calculate the probability that this number is divisible by 2 or by 3?